UNDERSTANDING BEHAVIORS

Learning Outcomes

- Sensations & sense organs
- Perceptions and principles
- Attention and concentration

SENSATION & PERCEPTION

SENSATION

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SENSATION

psychophysics

DEFINITIONS

- ***1. Sensation*** is the activation of the sense organs by a source of physical energy.(***stimulus***)
- 2. The process that occurs when special receptors in the sense organs are activated, allowing various forms of outside stimuli to become neural signals in the brain.
- A ***stimulus*** is any passing source of physical energy that produces a response in a sense organ. (Understanding Psychology, 2023)

Number of Senses

Basic senses	Sight
	Auditory
	Smell
	Touch
	Taste

- Pain
- Pressure
- Temperature
- Vibration

Sl. No.	Type of Sensation	Sense Organ	Senses	Knowledge gaining
1.	Visual Sensation	Eye	Sight	83%
2.	Auditory Sensation	Ear	Hear	11%
3.	Olfactory Sensation	Nose	Smell	3.5%
4.	Taste Sensation	Tongue	Taste	1.0%
5.	Tactual Sensation	Skin	Touch	1.5%

Elements of Sensation

Quality

Extensit
y

Duratio
n

Clarity

Intensity

PERCEPTION

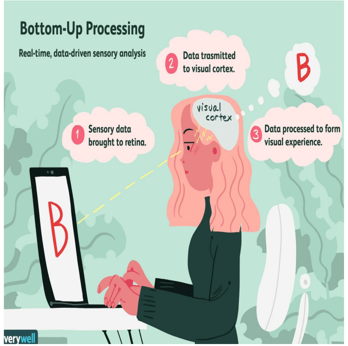
Principles of perception

PERCEPTION

DEFINITIONS

- ***1. Perception*** is the sorting out, interpretation, analysis, and integration of stimuli carried out by the sense organs and brain
- 2. The method in which the sensations experienced at any given moment are interpreted and organized in some meaningful fashion

Perception Processing's

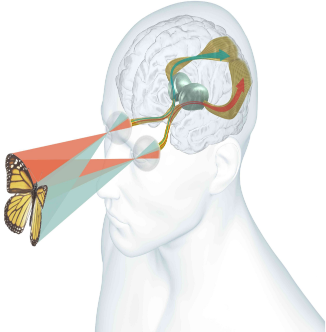


Bottom-Up Processing

- Data-driven
- Sensory receptors
- Do not rely on prior knowledge
- Unfamiliar stimuli.

Top-Down Processing

- Cognitive processes
- Memory
- Familiar situations
- knowledge, expectations



How the stimulus is detected

ABSOLUTE THRESHOLD

Humans can hear sounds down to 20

Hertz (vibrations) per second and up to 20,000 hertz.

This is a practical range because if your ears could sense tones below 20 hertz, you would hear the movements of your muscles!

An **absolute threshold** is the smallest intensity of a stimulus that must be present for it to be detected

Signal Detection Theory

Signal detection theory is a method of differentiating a person's ability to discriminate the presence and absence of a stimulus (or different stimulus intensities) from the criterion the person uses to make responses to those stimuli

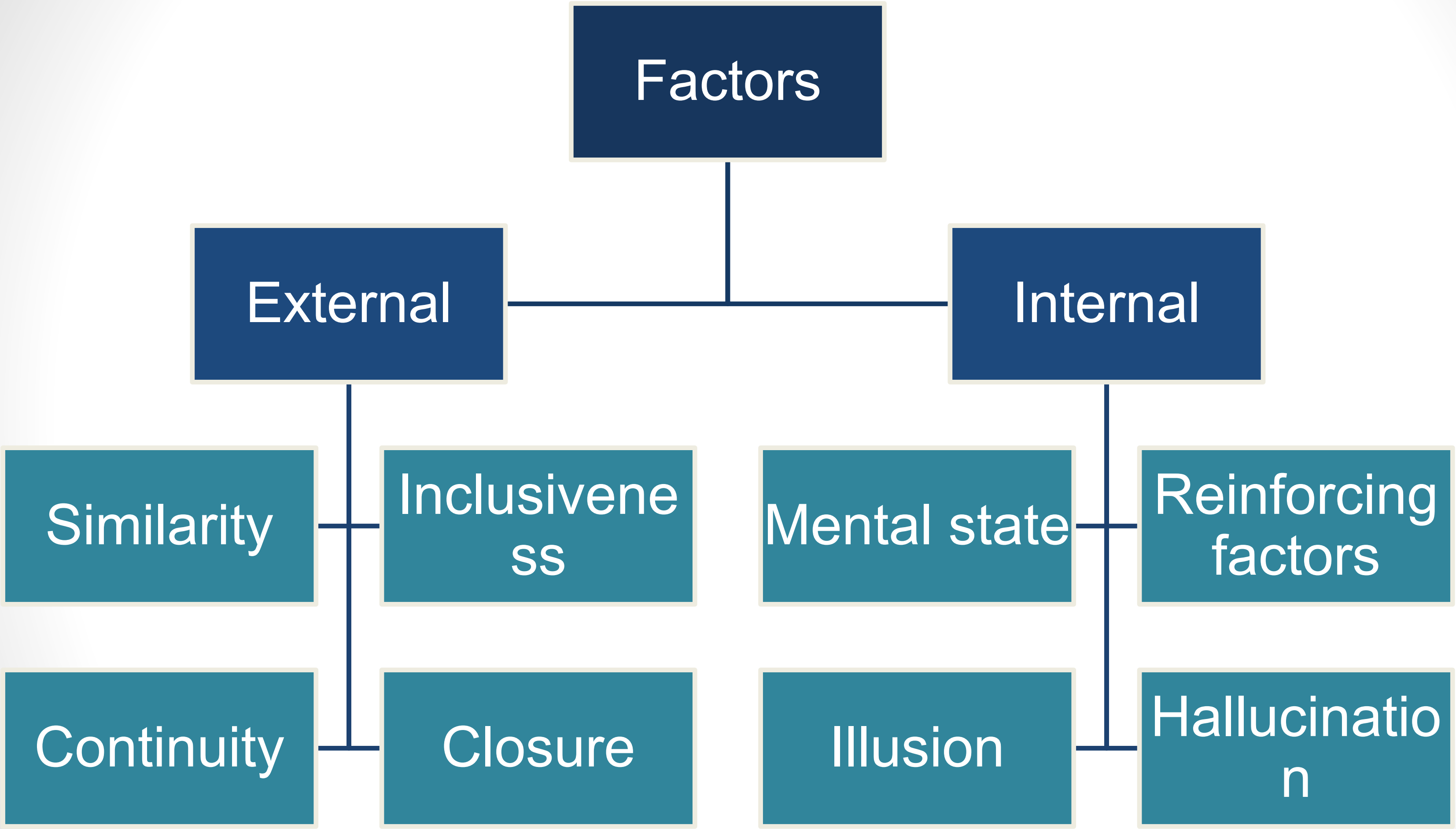
This refers to our attempt to focus on one particular stimulus and ignore the flood of information entering our senses.

Sensory Adaptation

Sensory Adaptation

- Tendency of sensory receptor cells to become less responsive to a stimulus that is unchanging.
- It is a reduction in sensitivity to a stimulus after constant exposure to it.
- While sensory adaptation reduces our awareness of a stimulus, it helps free up our attention and resources to attend to other stimuli in our environment.
- Receptors are no longer sending signals to the brain

FACTORS RELATED TO PERCEPTIONS


Factors
External
Internal
Similarity
Inclusiveness
Continuity
Closure
Mental state
Reinforcing factors
Illusion
Hallucination

Gestalt Principal’s

The External Factors

EXAMPLE
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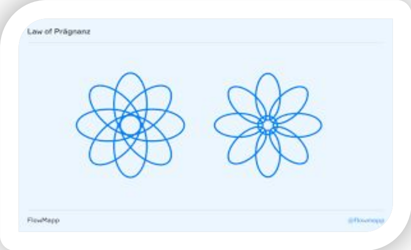
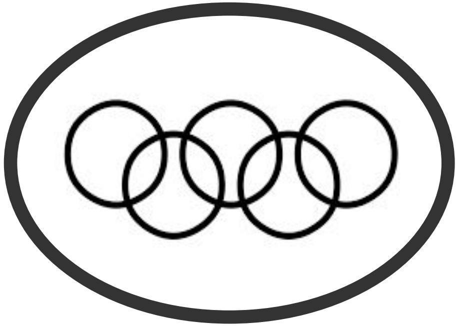
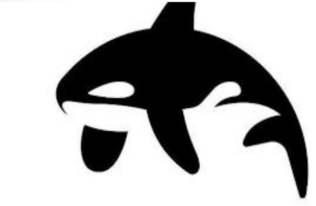


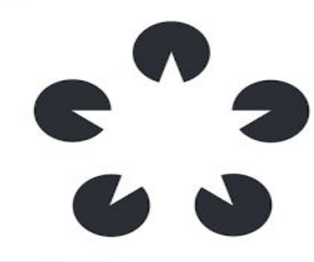


Developed by German psychologists, the Gestalt laws describe how humans perceive the world around them through a psychological process known as ***perceptual organization.***

In this process, the human mind will group stimuli (objects) into comprehensible patterns in order to readily determine their meaning

PERCEPTUAL CLOSURE





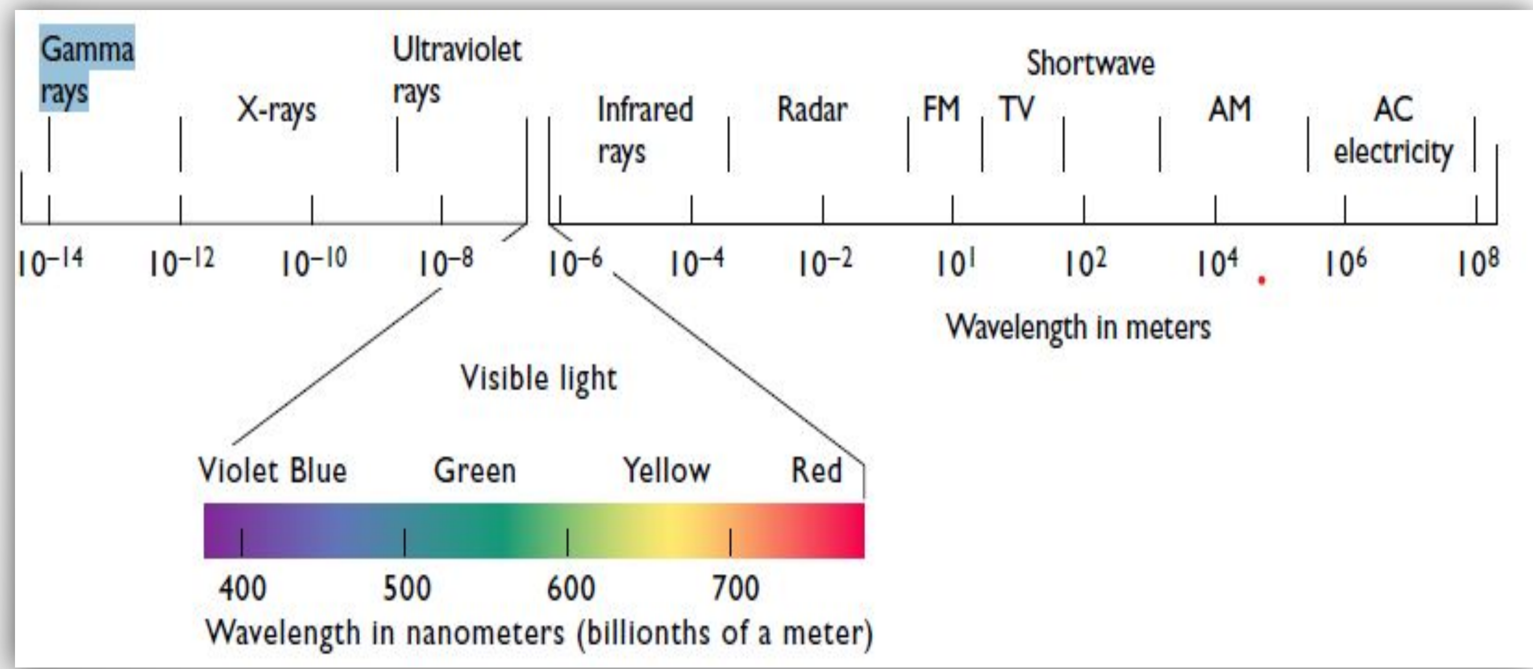
Gestalt psychologists believe that the brain tends to perceive forms and figures in their complete appearance despite the absence of one or more of their parts, either hidden or totally absent.

Gestalt psychologists claimed that when we receive sensations that form an incomplete or unfinished visual image or sound, we tend to overlook the incompleteness and perceive the image or sound as a complete or finished unit. This tendency to fill in the gaps is referred to as closure.

Assignment

- Principle of continuity
- Principle of inclusiveness
- Principle of similarity

Color – Vision & Color – Blindness



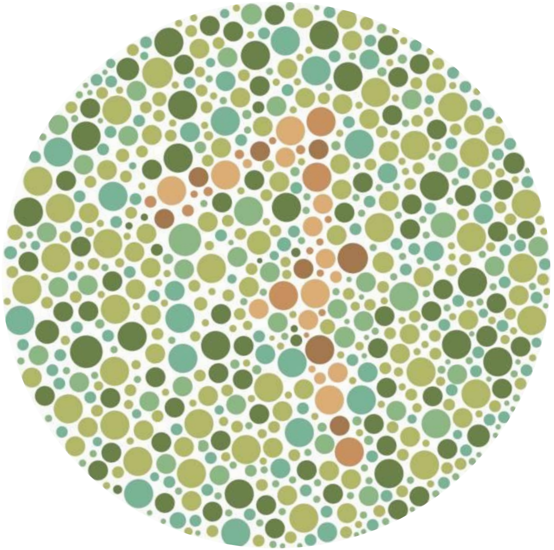
- Although the range of wavelengths to which humans are sensitive is relatively narrow, at least in comparison with the entire electromagnetic spectrum, the portion to which we are capable of responding allows us great flexibility in sensing the world.

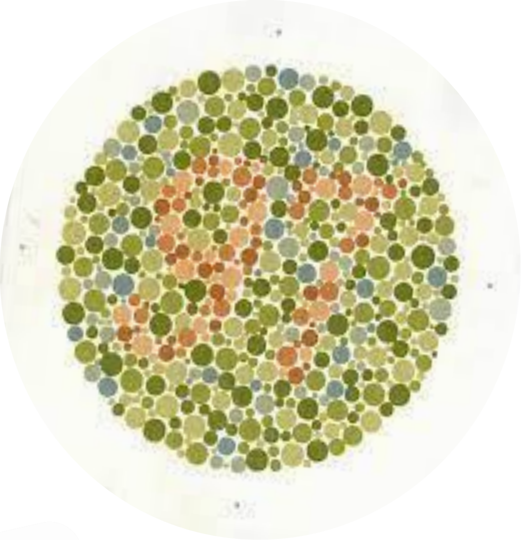
A person with normal color vision is capable of distinguishing no less than 7 million different colors

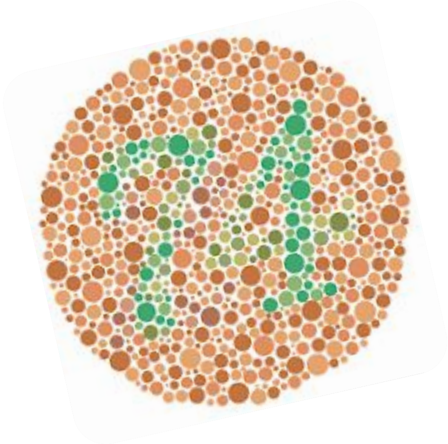
COLOR BLIND RATIO

7% of men

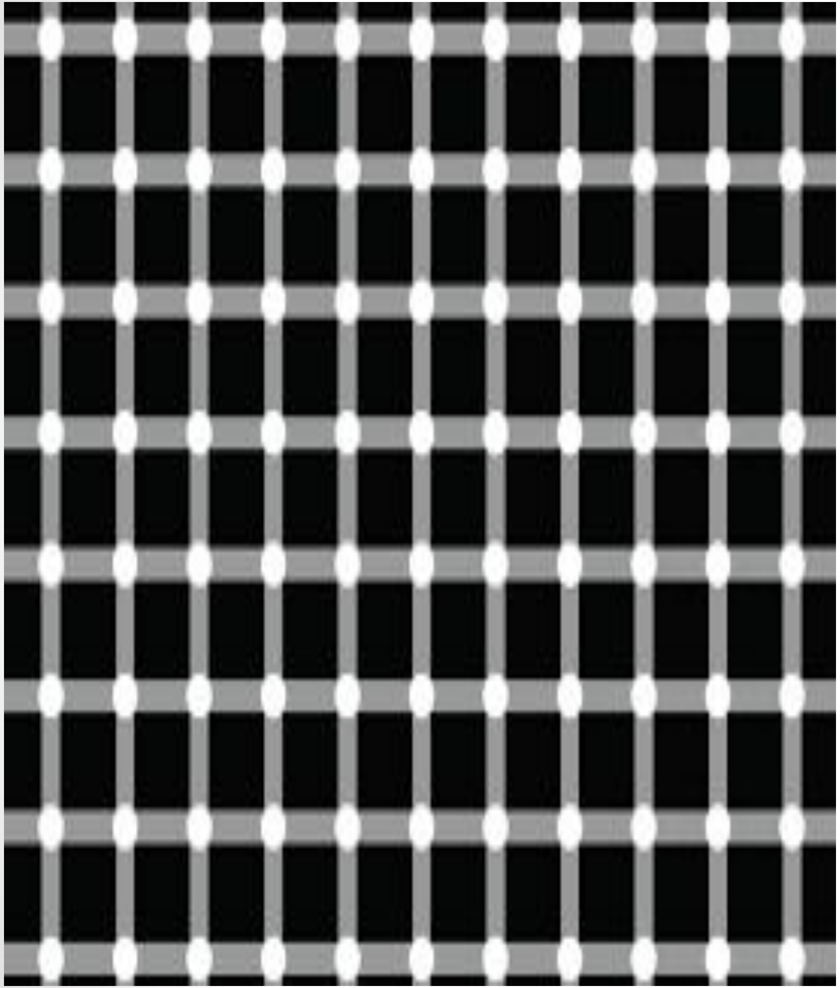
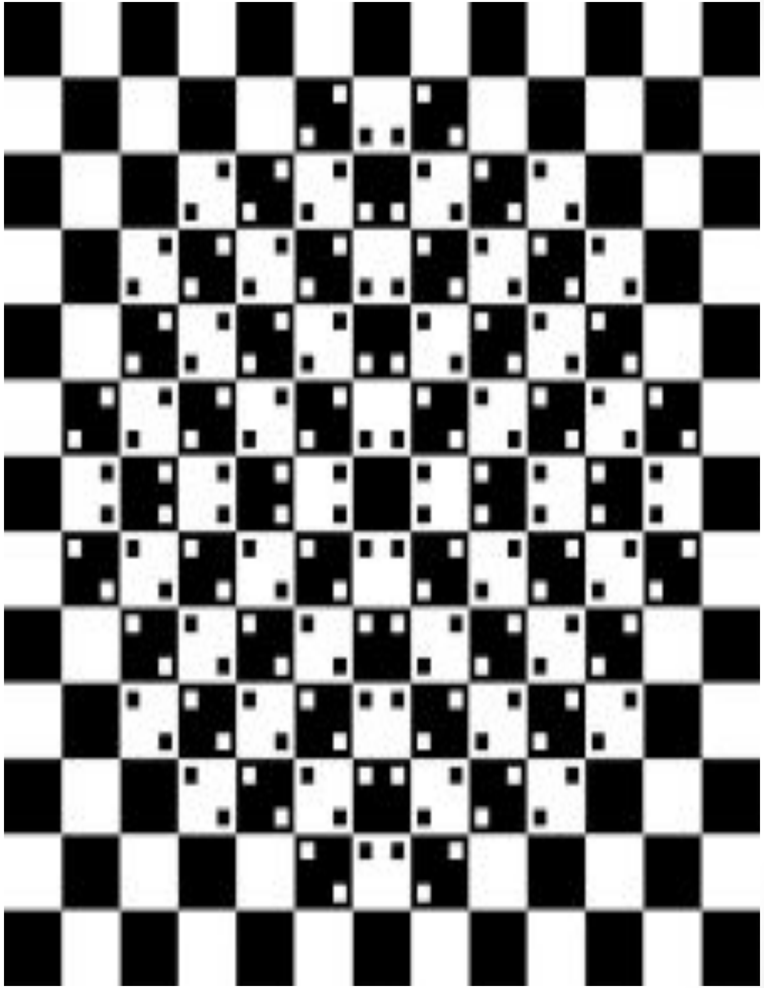
4% of women

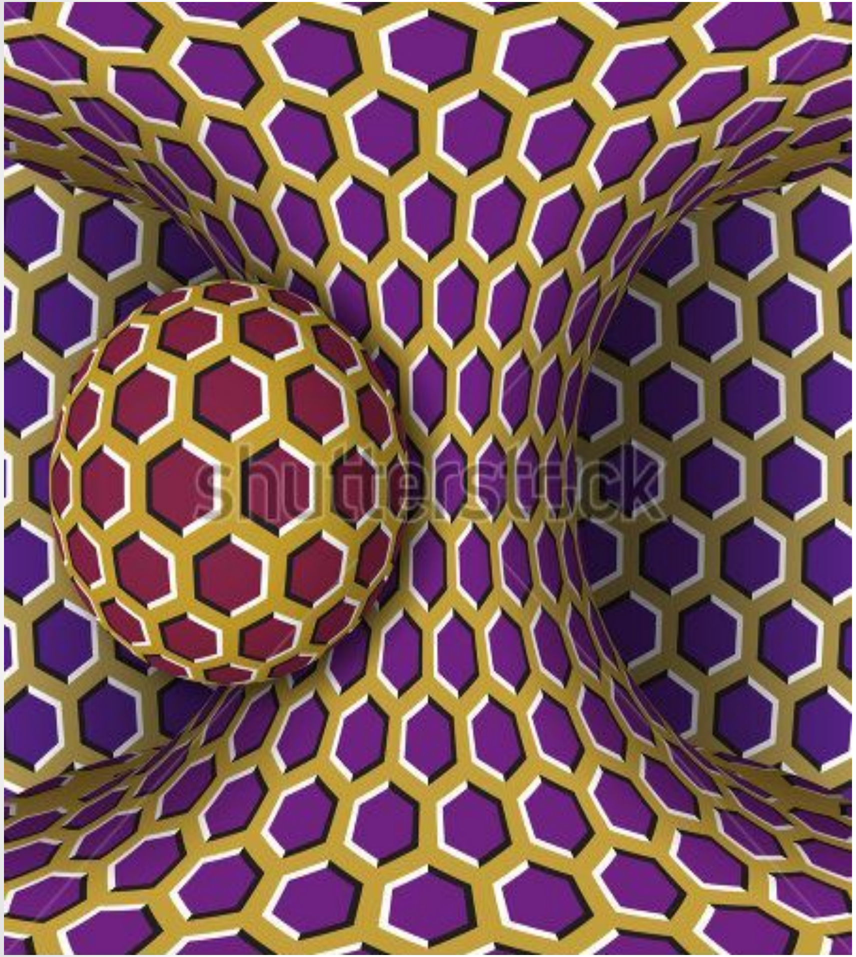






ILLUSION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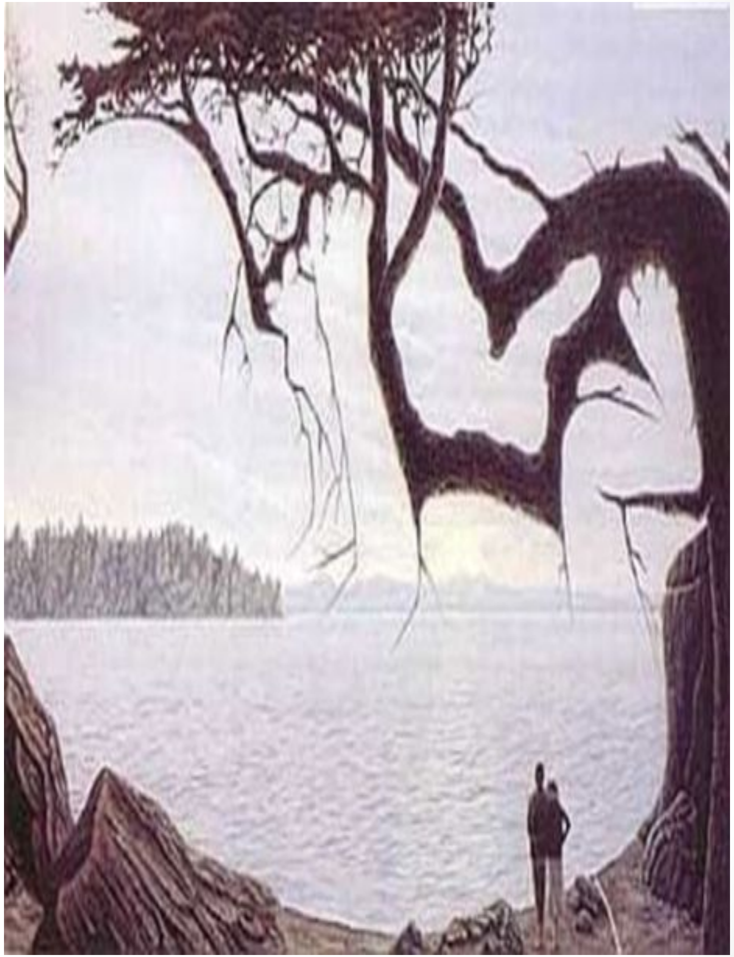
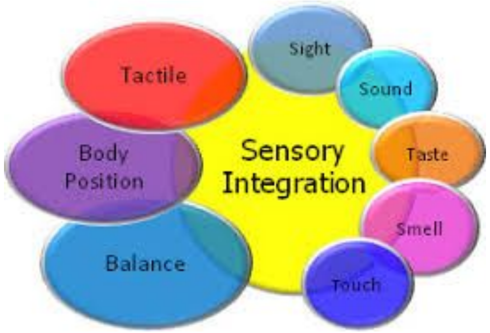


How Our Senses Interact?

We engage in multimodal perception, in which the brain collects the information from the individual sensory systems and integrates and coordinates it.

Moreover, despite the fact that very different sorts of stimuli activate our individual senses, they all react according to the same basic principles that we is cussed at the start of this chapter



PERCEPTUAL CONSTANCY
The phenomenon in which physical objects are perceived as unvarying and consistent despite changes in their appearance or in the physical environment .

THANK YOU ☺️